

REVIEW

Cholesterol glucosylation-based survival strategy in *Helicobacter pylori*

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Abstract

Helicobacter pylori is a major chronic health problem, infecting more than half of the population worldwide. *H. pylori* infection is linked with various clinical complications ranging from gastritis to gastric cancer. The resolution of gastritis and peptic ulcer appears to be linked with the eradication of *H. pylori*. However, resistance to antibiotics and eradication failure rates are reaching alarmingly high levels. This calls for urgent action in finding alternate methods for *H. pylori* eradication. Here, we discuss the recently identified mechanism of *H. pylori* known as cholesterol glucosylation, mediated by the enzyme cholesterol- α -glucosyltransferase, encoded by the gene *cgt*. Cholesterol glucosylation serves several functions that include promoting immune evasion, enhancing antibiotic resistance, maintaining the native helical morphology, and supporting functions of prominent virulence factors such as CagA and VacA. Consequently, strategies aiming at inhibition of the cholesterol glucosylation process have the potential to attenuate the potency of *H. pylori* infection and abrogate *H. pylori* immune evasion capabilities. Knockout of *H. pylori* *cgt* results in unsuccessful colonization and elimination by the host immune responses. Moreover, blocking cholesterol glucosylation can reverse antibiotic susceptibility in *H. pylori*. In this work, we review the main roles of cholesterol glucosylation in *H. pylori* and evaluate whether this mechanism can be targeted for the development of alternate methods for eradication of *H. pylori* infection.

KEYWORDS

antibiotic resistance, antibiotic susceptibility, cholesterol glucosylation, cholesteryl glucosides, drug development, *H. pylori*, *Helicobacter pylori*, immune evasion, lipid raft

1 | INTRODUCTION

H. pylori is one of the most prevalent pathogens that colonizes human gastric mucosa. It is assumed that more than half of the world's population is infected with *H. pylori*. The prevalence of *H. pylori* infection appears to be higher in low- and middle-income countries. The degree of *H. pylori* pathogenesis varies from one individual to another and ranges from asymptomatic infection to clinically overt diseases such as peptic ulcer, gastric atrophy, gastric mucosa-associated lymphoid tissue (MALT) lymphoma, and gastric cancer. Notably, *H. pylori* is categorized

as a class 1 carcinogen by the International Agency for Research on Cancer/World Health Organization, on the grounds that more than 85% of gastric cancer incidents were linked to *H. pylori* infection.¹ Gastric cancer is a major global health concern as every year it causes more than half a million deaths worldwide, making it one of the three major causes of cancer-related causalities.^{2,3} The development of *H. pylori* associated gastric cancer could be altogether prevented by eradication of the infection.⁴ This is, however, obfuscated by the increasing antimicrobial resistance rates in *H. pylori*. Resistance rates for certain antibiotics reached 50% in some countries, and the picture is

further complicated by the emergence of multi-drug-resistant strains.⁵

H. pylori employs several strategies to persist within the host. For instance, urease neutralizes the acidic pH of the stomach and the helical architecture of this gastric pathogen enables a swift escape through the viscoelastic gastric mucosa.^{6,7} In order to ensure its persistence, *H. pylori* also manipulates the host immune responses. Among the various immune evasion strategies, the prominent one that has been recently brought into light is the acquisition of host cholesterol and the conversion of it into cholesteryl glucosides (CGs) by the *H. pylori* enzyme cholesterol- α -glucosyltransferase (Cgt), encoded by the *cgt* gene.^{8,9} Consequently, several studies have investigated the role of cholesterol glucosylation in *H. pylori* infection and the authors acknowledge that it represents an important mechanism for the establishment of chronic infection. Notably, the abrogation of the cholesterol glucosylation results in ineffective or unsuccessful colonization of the host stomach and significantly attenuates the virulence potential of *H. pylori*.¹⁰⁻¹² In this literature review, we analyze all facets of cholesterol glucosylation and its role in *H. pylori* infection. We describe how cholesterol glucosylation and its interplay with other virulence mechanisms help evading the host immune responses to establish a chronic infection. We also discuss the results of studies on how the prevention of cholesterol glucosylation abrogates the colonization potential of *H. pylori* and enhances the clearance of *H. pylori* infection by the host immune response.

1.1 | Significance of cholesterol- α -glucosyltransferase, encoded by the *cgt*, in *H. pylori*

Cholesterol biosynthesis and its utilization are characteristics of eukaryotes, and such traits are not known in prokaryotes. Cholesterol is an abundant sterol in mammalian cells, more than 90% of which is present in the cell membranes, and is a vital component of the eukaryotic cell physiology. In addition to its role in eukaryotes, it is indispensable for some pathogens, such as *Helicobacter* sp, *Mycoplasma* sp, *Mycobacterium* sp, and *Borrelia burgdorferi*, that acquire cholesterol from the host cells.^{8,13-15} It has been originally observed that the high-cholesterol nutrients from sources such as egg yolk can attract *H. pylori*, which then uptakes cholesterol without changing the hydrophobicity of the cell.¹⁶ Later, Hirari et al⁸ demonstrated that the presence of CGs is indeed one characteristic feature of *Helicobacter* sp. However, *H. pylori* does not possess the genes required to synthesize cholesterol endogenously; instead, the bacterium evolved a mechanism to uptake cholesterol from the surrounding environment (membranes, gastric mucus, or dietary cholesterol) and incorporate it in its surface in the form of glucosylated cholesterol.¹⁷ As mentioned, *H. pylori* is not the only pathogen requiring cholesterol. *M. tuberculosis*, for example, utilizes cholesterol as a carbon source and to synthesize toxic polyketides, making it fundamental for its growth, persistence, and virulence inside of the host. *M. tuberculosis* harbors more than 80 genes involved in cholesterol import and catabolism.^{15,18} In case of *H. pylori*, the principal canalization of glucosylation of cholesterol

is carried out by Cgt enzyme, which pilots the synthesis of four characteristic CGs; cholesteryl- α -glucoside (α CG), cholesteryl-6'-O-phosphatidyl- α -glucoside (α CPG), cholesteryl-6'-O-acyl- α -glucoside (α CAG), and cholesteryl-6'-O-lysophosphatidyl- α -glucoside (lyso- α -CPG). It was noted that the deletion of the *cgt* leads to the loss of all CG variants.^{8,9} Interestingly, it was also observed that when *H. pylori* is grown in a hermetic tube having a gradient of cholesterol concentration, it can use its chemotactic ability to swim in the direction of highest concentration of cholesterol, and thrive there. Moreover, it was shown that *H. pylori* is associated with the cholesterol-rich membranes of epithelial cells and extracts the eukaryotic cholesterol from host gastric epithelial cells using the Cgt enzyme, which in turn compromises the lipid raft of the host cells.¹⁰

Using immunoelectron microscopy, it has been demonstrated that the synthesis of the Cgt enzyme occurs in the cytoplasm and it is subsequently exported to the inner membrane of the bacterial cell. Remarkably, only Cgt isolated from the membrane showed enzymatic activity. The other fractions of Cgt, including the cytoplasmic fraction, appeared to be inactive.¹⁹ Recently, the gene *hp0499* was identified as responsible for encoding the cholesteryl α -D-glucopyranoside acyltransferase (CGAT), which converts α CG into α CAG. CGAT enzyme is secreted and transported via outer membrane vesicles to the host cells for the direct synthesis of CAGs.²⁰ These pieces of evidence suggest that Cgt is a critical enzyme for the synthesis of cholesteryl glucosides in *H. pylori*.

1.2 | Cholesteryl glucosides are critical for cell membrane integrity in *H. pylori*

Some bacterial pathogens, such as *Borrelia*, *Proteus*, and *Mycoplasma* genera, can utilize cholesterol from the exogenous environment and incorporate it into their cell membranes. These pathogens possess the catalytic enzymes required for the conversion of host cholesterol into cholesterol glycolipid variants.²¹ For instance, *B. burgdorferi*, a causative agent of Lyme disease,²² can incorporate cholesteryl galactosides into its membrane, resulting in it comprising more than 36% of the bacterium's total glycolipids.^{23,24} The presence of CGs in the cell membrane is a characteristic feature of *Helicobacter* spp., although the content of CGs varies among different species.^{25,26} In addition to this, *H. pylori* CGs are linked to unique lipid chains that have 3-OH 14:0 and 3-OH C19:0 in its membrane, making it somehow similar to the membranes of archaea. This may also have a role in helping *H. pylori* to survive in the highly acidic environment of the stomach.²⁷ *H. pylori* α CAG is accompanied by variable unsaturated acyl chains such as α CAG (16:0), α CAG (18:0), α CAG (18:1), α CAG (18:2), and α CAG (20:4) that are involved in the formation of lipid rafts. *H. pylori* extracts these phospholipids from the gastric epithelial cells of the host for the biosynthesis of α CAG, which is required for promoting its lipid-raft formation.²⁸

Furthermore, phosphatidylethanolamine (PE) of *H. pylori* in the outer membrane binds to non-esterified cholesterol and myristic acid of the host (C14:0).²⁹ The *H. pylori* membrane also possess

flotillin-like protein HP0248/FLOT that is likely involved in the accumulation of cholesterol microdomains.³⁰

A crucial role of glycolipids is protecting bacteria from unfavorable conditions by altering the cell wall glycolipid content and consequently changing the cell shape in response to environmental stimuli and depending on the infection stage. *H. pylori* changes its morphology from spiral to coccoid in the decline phase of growth, or in response to various stress conditions. The morphology transition is accompanied by variation in the CG content: α CAG level increases in spiral form, while α CPG is sparsely present in the log phase. Subsequently, α CPG level increases during the coccoid conversion in the lag phase.³¹ Moreover, the absence of CGs increases the *H. pylori* membrane fluidity accompanied by distortion of typical *H. pylori* morphology and conversion into "c"-shaped cells.³² The lipid rafts in *H. pylori* are altered and modified according to the infection stages and in response to the environmental stress. Similarly, cholesterol-containing lipid raft-like microdomains in *B burgdorferi* alter their organization according to the transmission cycle of the spirochetes and to the exposure to a range of temperatures.³³ The membrane lipid rafts in *B burgdorferi* are similar to the eukaryotic lipid rafts. Cholesterol depletion in *B burgdorferi* and its replacement with eukaryotic lipids has been shown to cause changes in the morphology and membrane permeability according to the type of sterol that has been replaced.³⁴

H. pylori lacking CGs results in defective lipopolysaccharide (LPS) structures such as dephosphorylation of lipid A at the position-I with altered Lexis X (LeX) and Lexi Y (LeY) antigens.³⁵ In addition to this, the deletion of the *cgt* in *H. pylori* 76 strain disrupted the normal components of LPS resulting in the loss of O-antigens and the core LPS.³²

To summarize, *H. pylori* possesses membrane lipid raft maintained by CGs that help *H. pylori* to survive in harsh environment. *H. pylori* CGs are also remarkably important in maintaining the native helical morphology and typical LPS structure. Overall, CG lipid rafts are critical for the establishment of *H. pylori* chronic infection.

1.3 | Role of CGs in the proper functioning of prominent *H. pylori* virulence factors

The oncoprotein cytotoxin-associated gene A (CagA) is one of the most studied virulence factors of *H. pylori*.³⁶ The translocation of phosphorylated CagA into gastric epithelial cells occurs through the type IV secretion system (T4SS).³⁷ Once the CagA enters into AGS cells, it alters the morphology of the host cells resulting in distorted, elongated cells known as the "hummingbird phenotype".³⁸ Interestingly, CagA translocation in cholesterol-depleted AGS cells showed decreased hummingbird phenotype and IL-8 expression.¹² This is because CGs are required for the formation of lipid rafts in *H. pylori* and thus by depleting host cholesterol, delivery of CagA through T4SS is hampered.²⁸ CagA also induces an inflammatory response in epithelial cells. It is believed that T4SS is substantial for the induction of NF- κ B-dependent cell response, in

which peptidoglycans are delivered into epithelial cells via T4SS. Subsequently, the PGs are recognized by cytosolic nucleotide-binding oligomerization domain protein 1 (NOD1) downstream of the pathogen recognition molecule (PRM), and NOD1 activates the NF- κ B and IL-8 responses.^{39,40} CagL binds and activates the $\alpha 5 \beta 1$ integrin receptor in T4SS, which triggers the injection of CagA.⁴¹ Interestingly, the depletion of cholesterol from cells using methyl- β -cyclodextrin (M β CD) results in the disruption of $\alpha 5 \beta 1$ integrin receptors present in cholesterol-rich microdomains, which consequently leads to decrease in expression of NF- κ B and IL-8-dependent responses against *H. pylori*.⁴²

Another strategic *H. pylori* virulence factor is the vacuolating cytotoxin (VacA) that induces gastric epithelial cell erosion, vacuolation, and ulceration.⁴³ Cholesterol depletion from AGS cells by M β CD remarkably attenuates the entry of VacA into the cells and subsequently abrogates vacuolation.⁴⁴ Overall, cholesterol glucosylation is a key mechanism for the typical functioning of prominent virulence factors of *H. pylori* during infection. Any interference with the cholesterol glucosylation process of *H. pylori* remarkably attenuates the pathogenicity in *H. pylori*, at least in vitro (Figure 1).

1.4 | Role of cholesterol glucosylation in *H. pylori* immune evasion

Cholesterol glucosylation in *H. pylori* is one of the crucial strategies to evade the host immune response, interfering with bacterial internalization in macrophages and neutrophils. Briefly, *H. pylori* binds to macrophages through adhesins such as sialic acid-binding adhesin A SabA, a tight binding that also requires a functional T4SS.⁴⁵ *H. pylori* then avoids internalization through the atypical protein kinase C- ζ (PKC- ζ) class IA phosphoinositide 3-kinases (PI3K) complex pathway instead of the typical PKC α pathway. Moreover, *H. pylori* internalization is hampered by the alkaline environment generated by urease activity and by the inhibition of the fusion of phagosome-lysosomes by VacA, through recruiting and retaining coronin 1 that stimulates the formation of megasomes.^{46,47} Surprisingly, *cgt* knockout strains showed enhanced internalization with decreased arrests of phagosome maturation.⁴⁸ This enhanced internalization of *H. pylori* by macrophages is probably due to the alteration in T4SS structure and attenuation of the VacA function.¹² Hence, the absence of CGs disrupts the T4SS structure and attenuates the VacA function. This results in the phagosome maturation proceeding normally, ultimately leading to the pathogen clearance.⁴⁴ Interestingly, wild-type *H. pylori* showed delayed macrophage autophagy in comparison with a *cgt* knockout, and deletion of autophagy-related protein 12 (Atg12) reduced the survival of internalized *H. pylori* in macrophages and hindered autophagosome maturation.⁴⁹

It is known that interleukin-2 (IL-2) induces cell-cycle progression and proliferation of primary human T cells and natural killer cells.⁵⁰ VacA interferes with IL-2 function and efficiently blocks the proliferation of T cells by inducing a G1/S cell-cycle arrest.^{51,52} Mechanistically, VacA binds to the $\beta 2$ (CD18) integrin receptor of

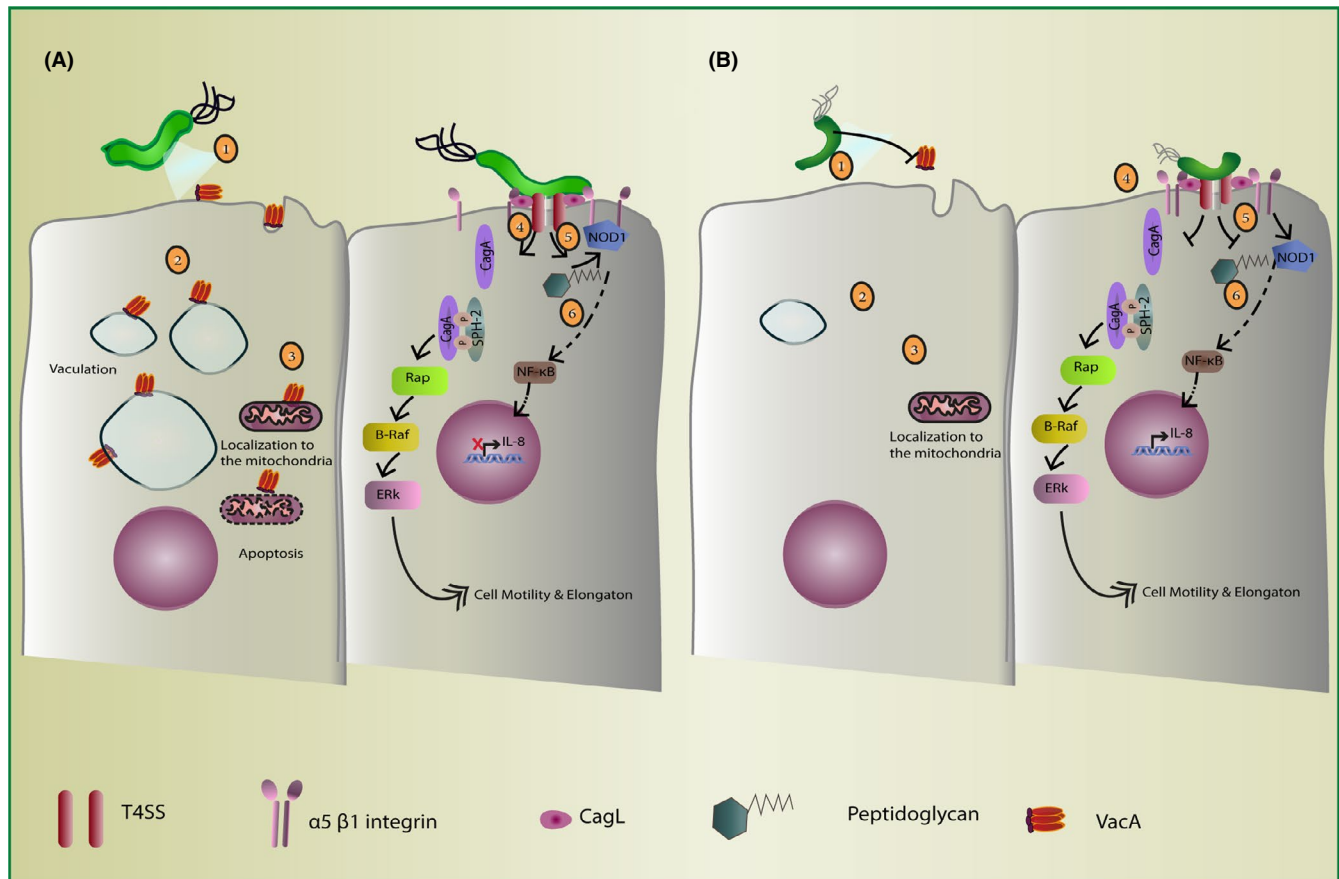


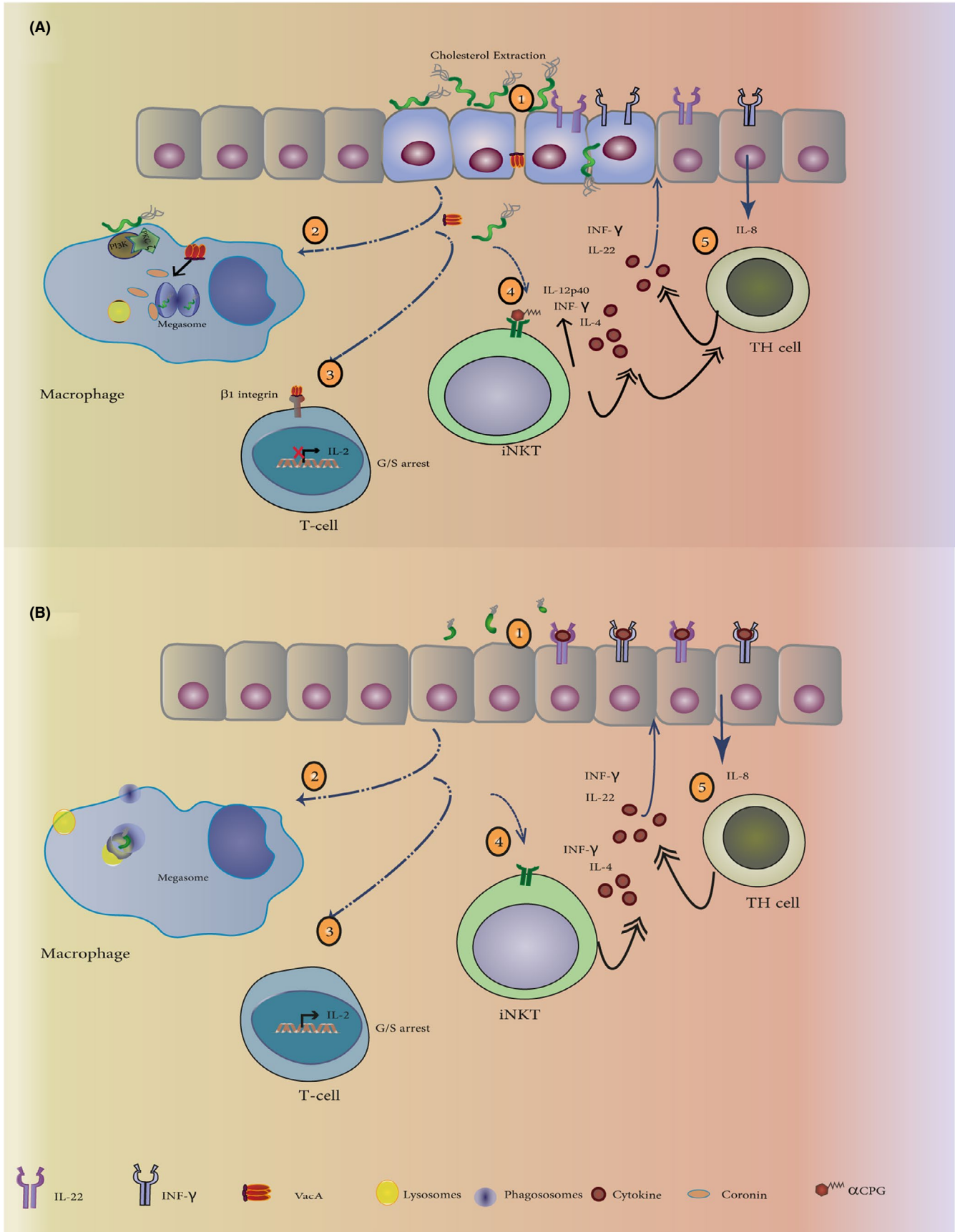
FIGURE 1 Interplay of cholesterol glucosylation with other virulence factors in *H. pylori*: (A) *H. pylori* secretes VacA that is internalized into gastric cells via endocytosis (1), which induces vacuolation by retaining the endosomes to NH₄ (2) and localizes to the mitochondria inducing apoptosis (3). *H. pylori* translocate CagA via T4SS system and induce gastric cell elongation (4) and the translocated PGs bind to NOD1, but due to the disrupted gastric cholesterol-rich microdomains, it prevents the assembly of $\alpha 5 \beta 1$ integrin that was activated via CagL (5) and fails to induce IL-8 secretion (6). (B) Deletion of *H. pylori* *cgt* attenuates the production of VacA (1) and blocks VacA-based vacuolation (2) and apoptosis (3). In addition, it attenuates the translocation of CagA via distorted T4SS (4) and enhances the binding of PG to NOD1 that is activated via $\alpha 5 \beta 1$ integrin, activated via CagL (5), and induces IL-8 secretion (6)

migrating human T lymphocytes and is then internalized by endocytosis exploiting the integrin recycling vesicles.⁵³ A *cgt* mutant of *H. pylori* fails to decrease the primary human CD4⁺ T-cell proliferation in comparison with the wild-type strain. This could be due to the attenuated VacA that exerts antiproliferative effects on T cells.⁵⁴

Cholesterol glucosylation protects *H. pylori* from phagocytosis in the mouse *in vivo* infection model. It has been demonstrated

that the *cgt* mutant strains get cleared from the mice stomach after 48 h as opposed to wild-type and reconstituted strains.¹⁰ This once more indicates that the cholesterol glucosylation process by *H. pylori* is required to evade the host immune response and colonize the host stomach. In addition, *H. pylori* infection recruits adaptive immune responses dominated by Th17 and Th1 cells, which subsequently induce IFN- γ , IL-4, and IL-17 expression.⁵⁵ The expression of INF- γ dominated by Th1 is critical to control the inflammatory

FIGURE 2 Cholesterol glucosylation in *H. pylori* is a key player in tailoring host immune response. (A) *H. pylori* extracts cholesterol from gastric cells and prevents assembly of IL-22 and IFN- γ receptors. This abrogates the subsequent downstream JAK/STAT1 signaling and decreases inflammation, despite a marked global T-based immune response (1). *H. pylori* inhibits the fusion of phagosome-lysosome by retaining the coronin, and delays the internalization by inducing PI3K-PKC- ζ complex (2). VacA binds to the $\beta 2$ (CD18) integrin interfering with IL-2-induced cell-cycle progression and proliferation of primary human T cells without affecting IL-2-dependent survival and efficiently blocking proliferation of T cells by inducing a G1/S cell-cycle arrest (3). α CPG triggers the induction of iNKT cells and induces the expression of IFN- γ , IL-2, and IL-12p40 (4). *H. pylori* infection induces secretion of IL-8 and recruits Th1 cells, which secrete IL-22 and IFN- γ (5). (B) Deletion of the *cgt* in *H. pylori* distorts the native helical morphology of the bacterium. Consequently, gastric cells maintain intact membranes and IL-22 successfully binds to IFN- γ (1). Deletion of the *cgt* enhances the internalization of the bacterium by macrophages (2) and fails to reduce the primary human CD4⁺ T-cell proliferation (3). Moreover, it reduces the activation of iNKT cells (4) and induces Th1-based response (5) [Correction added on Jan 11, 2020 after first online publication: Figure 1 was renumbered as Figure 2, and Figure 2 was renumbered as Figure 1.]



response against *H. pylori*.⁵⁶ IL-22 protects epithelial cells from early infections by inducing the production of antimicrobial peptides.⁵⁷ In addition, the lipid rafts in gastric epithelial cells display the IFN- γ receptors, IFNGR1 and IFNGR2, which are assembled in specialized cholesterol-rich microdomains.⁵⁸ Noticeably, *H. pylori* extracts cholesterol from gastric epithelial cells and disrupts the cholesterol-rich microdomains. Consequently, the assembly of IFN- γ and IL-22 receptors does not take place, abrogating the IFN- γ and IL-22-based MAPK (mitogen-activated protein kinase) and JAK/STAT downstream signaling pathways, normally crucial for an effective Th1 immune response. In contrast, the *cgt* mutant strain fails to disrupt the cholesterol-rich microdomains in the gastric epithelial cells.¹¹ Henceforth, efforts put in developing a Th1 response-based vaccine aimed at eradicating *H. pylori* could be ineffective due to the abrogation of Th1 cell-based immune responses by Cgt.

Interestingly, clinical *H. pylori* isolates show significant variation in Cgt activity. The level of Cgt expression appears to be significantly related to the degree of induction of invariant natural killer T (iNKT) cells. Induction of iNKT subsequently induces the expression of IFN- γ , IL-2, and IL-12p40 that are involved in *H. pylori* gastric atrophy. It was further observed that Th1-based response against *H. pylori* is significantly reduced in Ja182/2 (iNKT knockout) mice compared with WT mice. These observations led to the conclusion that α CPG triggers the induction of iNKT cells, a mechanism involved in *H. pylori* gastric atrophy.⁵⁹

Given the above pieces of evidence, it is clear that cholesterol glucosylation is critical in preventing *H. pylori* from phagocytosis by macrophage cells and in modulating T cell-dependent inflammatory responses. Moreover, disruption of cholesterol-rich microdomains by *H. pylori* around its niche (gastric epithelial cells) could prevent the assembly of IFN- γ , thereby abrogating the downstream signaling pathways. Abrogating the process of cholesterol glucosylation in *H. pylori* may render *H. pylori* vulnerable to the host immune responses, resulting in the clearance of *H. pylori* infection (Figure 2).

1.5 | Direct roles of cholesterol glucosylation in *H. pylori* pathogenicity

H. pylori is a pathogen with the ability to colonize a host for its entire lifetime. However, *H. pylori cgt* mutant strains fail to colonize mice stomach compared with wild-type strains.¹¹ Moreover, it is shown that *H. pylori cgt* mutant gets cleared from the infected mice within 48 h post-infection.¹⁰ This is most likely because CGs are required for maintaining the native helical morphology, which is essential for *H. pylori* to penetrate through viscous gastric mucin and colonize the gut.^{7,32}

Another study observed variation in Cgt activity in clinical isolates from 24 Japanese patients. Cgt activity correlated with the grade of gastric atrophy. Gastric atrophy degree was arbitrated based on *H. pylori* load, neutrophil infiltration, glandular atrophy, and intestinal metaplasia.⁵⁹

A strategy adopted by *H. pylori* to survive and persist in the host involves structural modifications in LPS lipid A to decrease toxicity and increased expression of O-antigens to mimic the structures of blood sialic acid-binding adhesin.^{60,61} This mimicking of host O-antigen components makes the bacteria "invisible" to immune cells.⁶² Moreover, it was shown that the inactivation of the O-antigens by deleting β -1,4-galactosyltransferase results in inefficient colonization compared with the parental strains.⁶³ Culturing *H. pylori* in the absence of cholesterol leads to the formation of aberrant LPS which also reduces the expression of Lewis X and Y antigens and subsequently the strain fails to colonize.³⁵ The absence of CGs leads to the loss of O-antigens and core LPS in *H. pylori*,³² and the secretion of CAG OMVs significantly enhances the lipid rafts clustering, gathers adhesion molecules (including Lewis antigens and integrins α 5, β 1), and promotes bacterial adhesion.²⁰ Taken together, these studies suggest that CGs are directly involved through several mechanisms in *H. pylori* pathogenesis.

1.6 | Cholesterol glucosylation as a firewall against antibiotics in *H. pylori*

WHO has listed *H. pylori* among the 16 antibiotic-resistant pathogens that pose the greatest threat to human health.⁶⁴ Currently, there are relatively few options in terms of antibiotics available for the eradication of *H. pylori* infection. Unlike other pathogens, eradication of *H. pylori* requires a treatment regimen of triple or quadruple therapy due to its presence in low pH conditions and being it often multi-drug-resistant.^{65,66} Interestingly, when *H. pylori* was grown in the absence of cholesterol, the bacteria have shown susceptibility to several antibiotics.⁶⁴ Cholesterol is believed to reduce antimicrobial peptide (AMP) affinity to cholesterol-rich lipid membranes present in bacteria.⁶⁸ In line with these observations, the deletion of the *cgt* renders the bacteria susceptible to antibiotics, and the resistance to antibiotics is restored in *cgt*-reconstituted strains. Recently, it has been demonstrated that multi-drug resistant strains of *H. pylori* have high concentrations of CAGs.²⁰ Similarly, it was observed that *H. pylori* resistance to bile salts and ceragenins is maintained in *cgt* mutants, although this is known to be dependent on cholesterol.⁶⁷

In conclusion, *H. pylori* possesses membrane lipid raft maintained by CGs that help to maintain cell wall integrity, which in turn leads to intrinsic resistance against some antibiotics.

1.7 | Cgt as a potential therapeutic target to eliminate *H. pylori* infection

The resistance to antibiotics is raising at an alarming rate, and drug discovery has gained a priority area in the antimicrobial resistance (AMR) era. *H. pylori* is a highly resistant pathogen, and its eradication is increasingly problematic. Numerous studies have investigated alternative regimens and therapies for eradication of *H. pylori*, but the failure rates are still very high.^{5,69-71} Therefore,

the development of a target-based-drug could be an alternative approach to decrease the *H. pylori* burden. It is not surprising that Cgt has been considered as a candidate drug target. Previously, it was reported that O-glycans, which are naturally present in gastric mucin, inhibit CG synthesis and thereby prevent the growth of *H. pylori*, especially in the lag phase.^{72,73} Azasugar inhibitors inhibit the activity of CGT. These include glucosidase inhibitors such as castanospermine and 1-deoxynojirimycin, which are the potential lead compounds for the development of therapeutic inhibitors of CGT enzyme. A high potency was achieved in CGT inhibition when UDP was added in conjunction with the inhibitors.⁷⁴ Further, studies on recombinant Cgt protein showed that it was inhibited competitively by UDP-glucose, whereas it showed mixed inhibition responses in the presence of both cholesterol and UDP-glucose 2e.⁷⁵ Furthermore, it was shown that the clinically used drug amiodarone was a potent inhibitor of CGT and consequently reduced bacterial adhesion.²⁰ The crystal structure of the catalytic domain of Cgt is already determined and deposited in PDB with accession number: RCSP PDB (3QHP).⁷⁶ Overall, efforts toward the development of drugs that aim to inhibit *H. pylori* cholesterol glucosylation could hold strong potential for success towards decreasing the *H. pylori* infection burden.

2 | SUMMARY AND CONCLUSION

H. pylori extracts host cholesterol, and before incorporating it into its surface, it has to glucosylate it via Cgt. CGs are involved in the maintenance of membrane lipid rafts, which support *H. pylori* to survive in harsh environments. *H. pylori* CG lipid-raft microdomains are crucial for the maintenance of the native helical morphology and

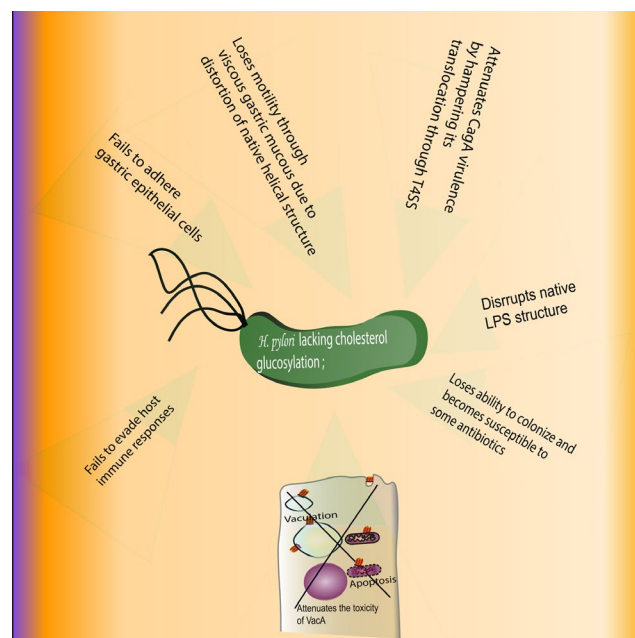


FIGURE 3 Schematic diagram highlighting the main consequences of abrogating the cholesterol glucosylation on the pathogenesis of *H. pylori* infection

the embedded typical LPS structure. The absence of CG lipid-raft microdomains could distort the *H. pylori* membrane structure and consequently disrupt the translocation of CagA via T4SS.

In addition, cholesterol glucosylation helps to maintain cell wall integrity and acts as a firewall against antimicrobial agents, which in turn leads to increased resistance to antibiotics.

Cholesterol glucosylation plays also a role in escaping both innate and adaptive immune responses. Cholesterol glucosylation

TABLE 1 Principal roles of cholesterol glucosylation in *H. pylori* infection

Major functions	The key strategies	References
1. Maintenance of membrane structure and function	- Regulates membrane integrity and permeability - Preserves the native structure of LPS - Support Lewis antigen presence on the surface of the cell wall	(Hildebrandt and McGee, 2009; Qaria et al, 2018)
2. Required for optimal functioning of virulence factors	- CG lipid raft microdomains support T4SS structure - Enhances the translocation of CagA - Involved in the secretion of VacA	(Jan et al, 2016; Lai et al, 2008; Patel et al, 2002)
3. Evasion of immune responses	- Extracts gastric cell membrane cholesterol and prevents cytokine assembly - Delays macrophage internalization - Blocks phagolysosome formation - Arrests T-cell cycle - Induces iNKT cells which results in gastric atrophy	(Beigier-Bompadre et al, 2011; Du et al, 2016; Ito et al, 2013; Morey et al, 2018; Wunder et al, 2006)
4. Contribution to <i>H. pylori</i> pathogenicity	<i>H. pylori</i> strain pathogenicity correlates with the activity level of Cgt - Maintains the native helical morphology and motility - CAG OMVs enhance the adhesion of <i>H. pylori</i> to gastric cells - Required for functional LPS and Lewis antigens	(Appelmek et al, 1998; Ito et al, 2013; Jan et al, 2020; Logan et al, 2000)
5. Resistance to antibiotics	- Reduces the affinity of AMP <i>H. pylori</i> antibiotic resistance level is correlated with Cgt activity	(Brender et al, 2012; Jan et al, 2020; Qaria et al, 2018; Trainor et al, 2011)

interferes with VacA function and, subsequently, delays the internalization of *H. pylori* by macrophages, and arrests the formation of the phagolysosome. In addition, cholesterol glucosylation inhibits T-cell proliferation and prevents T cell-based cytokine assembly at the membrane of the gastric epithelial cells.

Overall cholesterol glucosylation significantly contributes to the *H. pylori* pathogenicity, as the degree of gastric atrophy correlates with the activity of Cgt (Figure 3 & Table 1).

More than half of the world's population is infected with *H. pylori*, and around 10%–20% of these are at risk of overt clinical diseases for which there are currently no preventive interventions, and eradication failure rates are increasing. Discovery efforts⁷⁷ aiming at alternative treatment methods based on inactivation of cholesterol glucosylation in *H. pylori* could lead to the development of molecules that need not to kill bacteria directly, but rather to render *H. pylori* susceptible to clearance by the host immune system.

CONFLICT OF INTEREST

None.

DATA AVAILABILITY STATEMENT

Data openly available in a public repository that issues datasets with DOIs.

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